

CpE-494: Computer System Engineering

Project Abstract

Clevart: Adaptive In-Store Shopping Guidance System

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Signature Page

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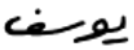
- I did my share of the work, and I have a general understanding of the contents of the assignment - I declare that this assignment is my own original work, except where the original source is cited. Name Signature Ayed Sultan Al-Mutairi 
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Needs

Supermarket customers can spend so much time inside the store, moving between aisles, managing shopping lists, and unplanned spending during a shopping trip, so finding products efficiently inside a physical store is still a real problem for shoppers, and a smart shopping cart that guides them could address it. Our survey of 49 respondents shows the problem clearly. The survey was conducted in October 2026 using Google Forms and was distributed through several group chats, producing 49 responses. About 61% reported often or very often having trouble finding an item (42.9% often, 18.4% very often). Nearly a third (30.6%) said they search for three minutes or more before giving up or asking for help. About half (only 30 participants out of the total 49 answer this question) (53.3%) walk back to an aisle they have already passed either every trip (30%) or most trips (23.3%). The impact on experience is evident as well: 44.9% reported feeling frustrated or angry when unable to locate an item, while 38.8% stated they might feel the same.

Most product discovery methods used today are manual. Of those surveyed (only 29 participants out of the total 49 answer this question), 48.3% just walk around and look, while 79.6% approach employees for assistance. Few make use of signs, apps, or store maps. This is important because 78.1% of respondents utilize cooperative societies (Jam'eya) (only 32 participants out of the total 49 answer this question) the most, and customers who visit unfamiliar stores (only 32 participants out of the total 49 answer this question) (40.6% only a few times a year) cannot rely on understanding the layout. There is a great need for a solution: A cart-mounted screen that displayed a live route to the items on their list was scored 5 out of 5 by 69% and 4 or 5 by 79.3% (only 29 participants out of the total 49 answer this question).

The research shows that going farther from their planned path led to a considerable rise in unexpected spending up to 57% (Hui et al., 2013) [1]. The study shows big supports this need where it's stated in section 4 "Decisions on how to guide consumers during the search process are important because consumers search relatively little and are likely to only discover and consider products that the retailer makes salient. " (Honka et al., 2024) [2]. Together with our survey, this indicates that a navigation cart can help shoppers by saving time and reducing frustration.

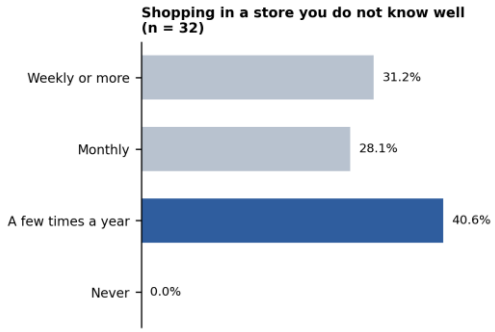


Figure 1: How often do you shop in a store you do not know well (new or rarely visited)?

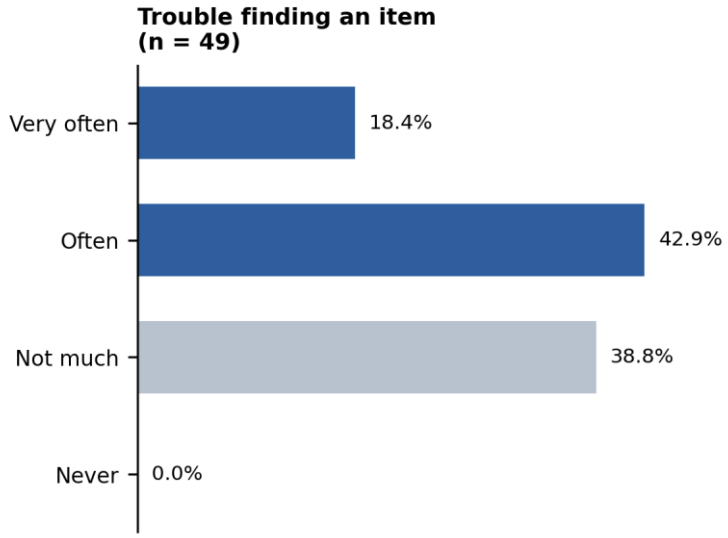


Figure 2: On a typical shopping trip, how often do you have trouble finding an item?

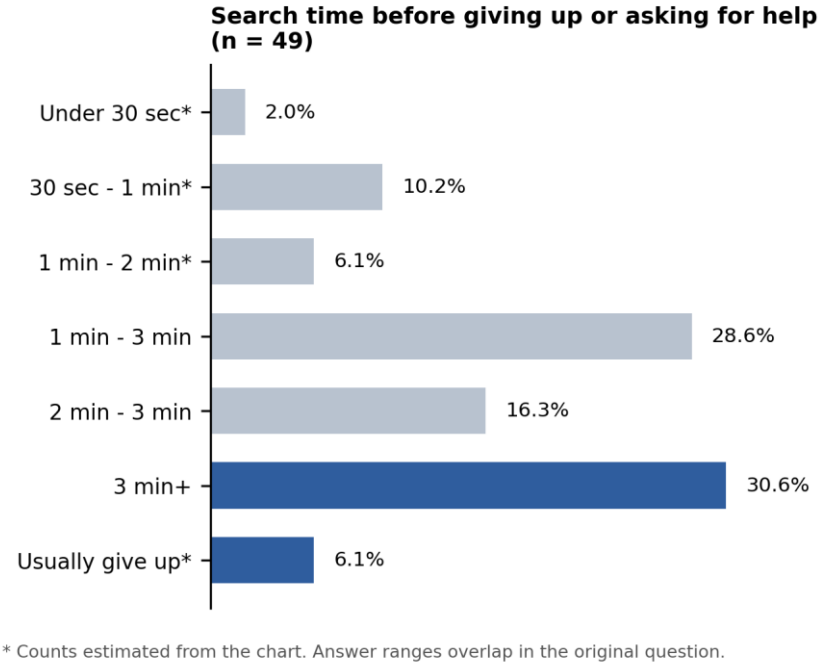


Figure 3: When you can't find an item, how long do you usually search before you give up or ask for help?

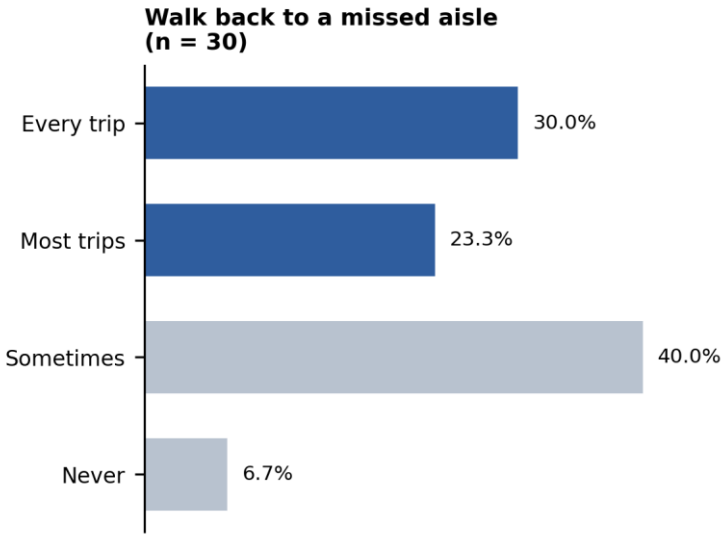


Figure 4: How often do you walk back to an aisle you already passed to get a missed item?

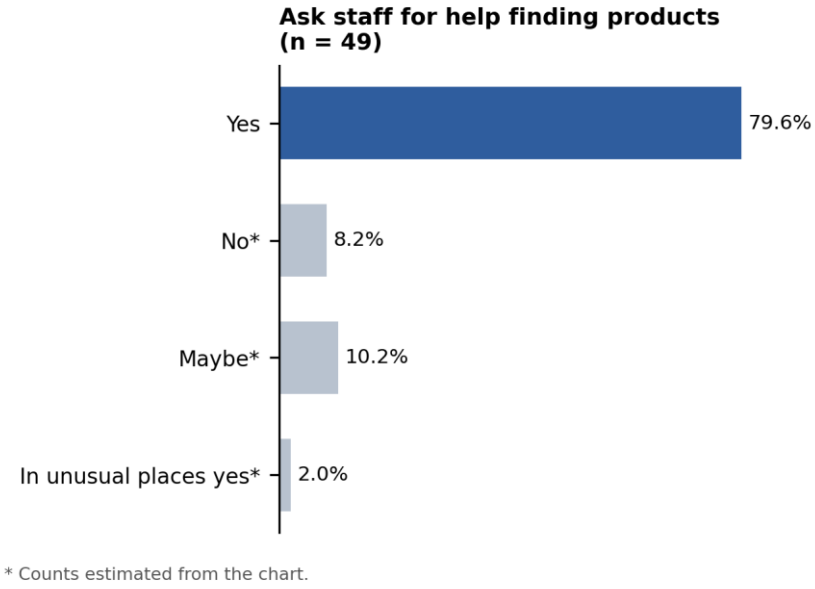


Figure 5: Do you ask staff for help finding products?

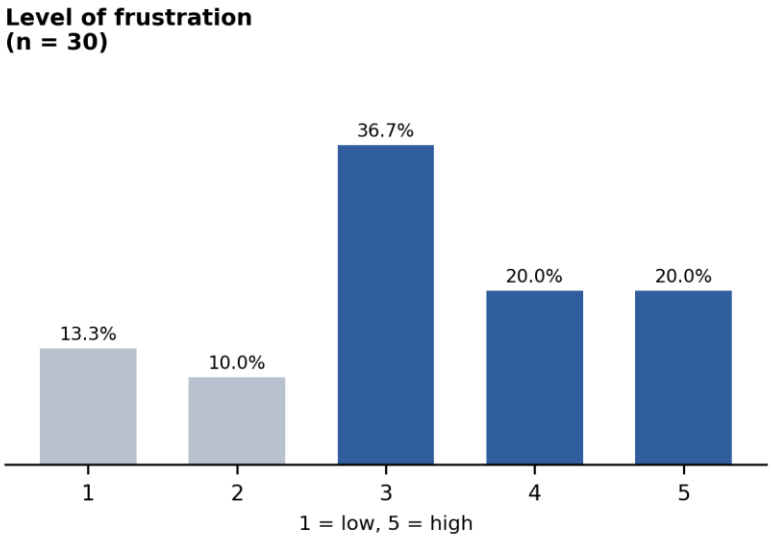


Figure 6: How frustrated do you feel when you can't find an item?

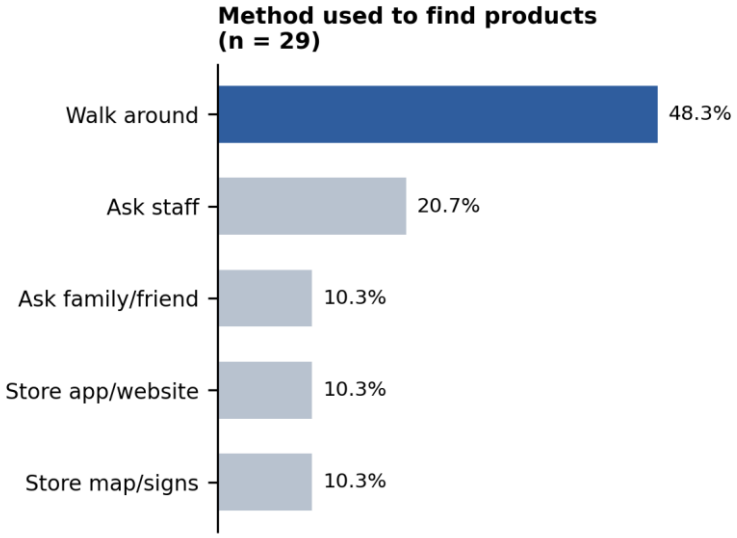


Figure 7: Which of these do you use to find products?

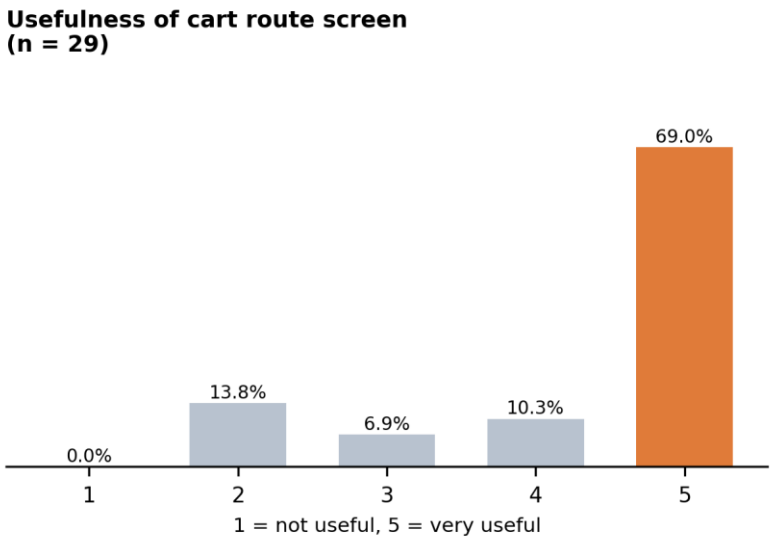


Figure 8: A screen on the shopping cart shows the route to the items on your list and updates as you shop. How useful would this be?

Objective

The primary objective of Clevart is to reduce product-search time and unnecessary walking in supermarkets through adaptive in-store guidance. The system aims to help shoppers locate products, avoid repeated trips between aisles, and complete their shopping lists with less dependence on store employees, particularly in unfamiliar stores.

Clevart will be developed as a **clip-on cart-mounted unit** with a display and an embedded controller. It will use a digital store map linked to a database containing product locations and prices. Store staff will verify and update product locations when items are moved. BLE beacons and camera-based markers will be evaluated during prototyping to select a suitable method for estimating the cart's indoor position. Using this position and the remaining shopping-list items, the system will display an efficient route and recalculate it when items are added, missed, removed, or collected, or when the shopper deviates from the planned route.

Purchased items will be tracked using **barcode or QR scanning** and matched with the product database to update the shopping list and running total. Computer vision may also be evaluated as an alternative or supporting method during prototyping. These supporting features will help shoppers monitor spending and reduce repeated manual checking.

The navigation function will be evaluated against shopping with a static store map by comparing **product-search time, walking distance, number of backtracking events, and the number of requested items successfully found**. Testing will include changes to shopping lists and product locations to assess the system's ability to adapt.

The minimum working prototype will include one clip-on cart unit, one mapped test area, a limited set of products, one indoor-positioning method, and adaptive routing; spending tracking, shopping history, and computer vision are optional features.

Clevart investigates dynamic multi-item route recalculation under changing shopping-list and shopper-position conditions in a lightweight clip-on cart system.

Background and Technology Survey

Background

Smart shopping systems have been developed to improve different stages of the supermarket shopping experience, including shopping-list management, product search, in-store navigation, purchase tracking, and checkout. These systems use technologies such as product databases, digital store maps, indoor positioning, computer vision, sensors, barcode or RFID identification, and mobile or cart-mounted interfaces. Although these technologies can reduce some shopping difficulties, effective product-finding guidance depends on knowing both the shopper's current location and the current location of products inside the store.

Research has specifically investigated product search and navigation. SmartMart connected a mobile search interface with a store-item database and floor map, allowing shoppers to search for products and view their locations. The work also identified changing product locations and maintaining accurate location information as practical challenges [3]. GroFin combined a digital shopping list with map-based contextual guidance. Its field evaluation reported reductions in shopping time and route length, showing that list-aware guidance can make supermarket navigation more efficient [4]. The Context-Aware Shopping Trolley (CAST) instead integrated assistance into the shopping trolley itself, and its evaluation reported easier product finding and shorter travel distance compared with a traditional trolley [5].

More recent research further demonstrates the importance of indoor localization and accurate product guidance. Sakman et al. developed a personalized shopping workflow using product search, indoor positioning, and electronic shelf labels. Their feasibility study used beacon-based localization and reported approximately 50–100 cm localization accuracy in the test environment, while also identifying deployment cost, calibration, and technology integration as practical concerns [6]. Picinin et al. proposed a hybrid supermarket-navigation system using augmented reality, computer vision for shelf-level recognition, and QR-code markers for spatial initialization. Their evaluation demonstrated world alignment and shelf detection while addressing BLE signal instability and image-processing latency [7].

A more hardware-intensive approach was presented by Liang et al. in 2025. Their smart shopping system combined Ultra-Wideband (UWB) positioning with Mask R-CNN and SLAM for localization and autonomous navigation. The study reported that

its MU-SLAM method reduced trajectory error by about 92% relative to ORB-SLAM3 on a public indoor dataset with simulated UWB data. In a 100-participant comparison against a conventional trolley with staffed checkout, the full system reduced average shopping time by 33.15% compared with a conventional workflow [8]. However, this approach requires substantially more sensing, computation, and store infrastructure than a lightweight shopping-assistance system.

Together, these studies indicate that navigation can reduce search effort and walking distance, but they also expose several engineering challenges relevant to Clevart: maintaining accurate product-location data, locating the shopper reliably indoors, adapting guidance when shopping conditions change, and avoiding excessive interaction while the user is moving through the store.

Enabling Technologies

Clevart is proposed as an adaptive in-store shopping guidance system that can be integrated with standard supermarket carts rather than requiring a completely autonomous robotic cart. The current preferred architecture is a **clip-on cart unit with a simple cart-mounted interface**, supported by a software application and store database. The mobile component can be used for preparing shopping lists, accessing purchase history, and account-related functions, while the cart interface can provide navigation during the shopping trip with minimal interaction.

The initial prototype is expected to use an ESP32-based embedded controller together with a cart-mounted Android tablet for the user interface. The tablet's built-in camera can support barcode or QR scanning, while the ESP32 can support BLE communication and interaction with the positioning system. A rechargeable power bank or battery pack will supply power to the cart-mounted electronics. These components are tentative and may be adjusted during prototyping based on cost, power consumption, and positioning requirements.

Indoor positioning is required because GPS is unreliable inside buildings. Possible technologies include Bluetooth Low Energy (BLE) beacons, Ultra-Wideband (UWB), and camera or QR-marker-based localization. For an initial Clevart prototype, **BLE-based positioning is a practical starting option because of its relatively low deployment cost and compatibility with common devices**. QR or visual reference markers may also be used at selected locations to correct or confirm position. For the minimum prototype, Clevart will target aisle-level positioning accuracy, which is sufficient to distinguish the shopper's current aisle and support route guidance between product locations. If BLE estimates become too inaccurate or ambiguous between nearby aisles, QR or visual markers placed at known reference points can be used to confirm and correct the cart's estimated position. UWB provides greater positioning accuracy and is therefore a possible alternative, but it requires additional infrastructure and hardware. Recent supermarket-navigation research has demonstrated both beacon-based approaches [6] and hybrid visual/marker approaches [7], while UWB has also been successfully integrated into more complex smart-cart navigation systems [8].

Product locations would be stored in a **store-managed product and location database**, where products are associated with aisle or shelf locations. Store staff or an administrator would update these records when products are relocated. This is important because previous research has identified changing product locations as a major practical problem for supermarket navigation systems [3].

If a shopper scans a product at a location that differs significantly from its stored database location, Clevart will flag the product-location record as potentially outdated. The cart's estimated position at the time of the scan can be used as supporting information, while the updated location will be verified by store staff before replacing the existing database record. This prevents a single inaccurate position estimate from automatically introducing incorrect product-location data.

For the prototype, purchased items will be scanned using the cart-mounted Android tablet's built-in camera with Google ML Kit Barcode Scanning. The detected barcode or QR code will be matched with the product database to update the shopping list and running total. This avoids the need for a separate dedicated scanner in the minimum prototype, while a dedicated scanner module can remain a future alternative if testing shows it is needed.

The supermarket layout can be represented as a graph containing aisles, intersections, and product locations. For multiple shopping-list items, the system will first determine an efficient visiting order using a nearest-neighbour heuristic, with 2-opt used to improve the route when practical. Dijkstra's algorithm or A* will then calculate the shortest path between consecutive product locations. The remaining route will be recalculated when an item is added, removed, missed, or marked as found, or when the shopper deviates from the planned route beyond a predefined distance threshold. This adaptive routing process is intended to reduce unnecessary backtracking while remaining practical for real-time use.

Existing Systems and Related Work

1. Coles Smart Cart

The Coles Smart Cart uses AI, sensors, and built-in scales to identify and track products placed in the cart. It provides customers with information about their shopping basket, store navigation features, and in-cart payment [9]. These capabilities demonstrate the value of combining shopping information and cart interaction. However, the cited source emphasizes product tracking, maps, and checkout and does not establish adaptive shopping-list rerouting when the shopper misses, adds, or changes an item.

2. Intermarché Smart Cart

The Intermarché system uses a clip-on Shopic device to upgrade an ordinary shopping cart. Computer vision is used for product recognition, and the system supports

spending tracking, promotions, loyalty information, and checkout [10]. Its clip-on architecture is particularly relevant to Clevart because it demonstrates that existing carts can be enhanced without replacing the entire trolley. The cited description, however, does not establish adaptive shopping-list-based route recalculation.

3. Woolworths Scan&Go

The Woolworths Scan&Go provides a cart-mounted device and integrated scanner so customers can scan and pack products during the shopping trip [11]. This reduces checkout-related effort and demonstrates how item tracking can be integrated directly into a cart. Woolworths' published information describes list and search capabilities as developing features, while adaptive shopping-list route recalculation is not established in the cited material.

4. SmartMart

SmartMart is a mobile shopping-assistance system that combines an Android application, an information-retrieval database, a digital store map, and RFID-based indoor positioning. Its approach allows store items to automatically register and update their location information so shoppers can search for products and view them on the store floor plan [3]. The study demonstrated the feasibility of product search and dynamic item localization, while in-store navigation and shortest-path calculation for multiple items were identified as future development. Clevart differs by focusing on adaptive route guidance that responds when the shopper's list, position, or remaining route changes during the trip.

5. GroFin

GroFin combines a digital shopping list with map-based contextual notifications and proximity information [4]. Its field evaluation reported reduced shopping time and optimized shopping paths while preserving the shopper's cognitive effort. This provides direct evidence that list-aware guidance can improve supermarket-shopping efficiency. Clevart extends this direction by focusing on **adaptive route recalculation** when items are added, missed, removed, or otherwise change during the shopping trip.

6. Context-Aware Shopping Trolley (CAST)

CAST integrates contextual assistance directly into a shopping trolley rather than relying only on a handheld device. Its evaluation reported that participants found products more easily and travelled shorter distances than users of a conventional trolley [5]. This supports the idea that assistance can be integrated into the cart to reduce interaction burden. Clevart differs by focusing specifically on product-location reliability and route adaptation rather than general context-aware trolley interaction.

7. Techno M-Cart

Techno M-Cart is a Kuwait University project centered on a robotic shopping cart using sensors, hand-gesture control, obstacle detection, and a mobile application [12]. Its main navigation function concerns autonomous cart movement and obstacle avoidance. Clevart differs by focusing on product-location guidance and adaptive shopping-list routing rather than robotic cart movement.

Table 1: Comparison of Existing Systems

System	Technical Approach	Navigation/Product Finding	Purchase Tracking	Strength	Limitation
SmartMart [3]	Android app + RFID indoor positioning + product database + digital floor map	Product search and mapping; shortest-path navigation proposed as future work	Limited	Automatically registers and updates product locations	Adaptive multi-item route guidance was not demonstrated
GroFin [4]	Smartphone + map-based list + contextual notifications	Context-aware guidance	Shopping-list management	Reduced shopping time and route length	Does not establish Clevart-style continuous adaptive rerouting
CAST [5]	Context-aware instrumented trolley	In-trip product-finding guidance	Limited	Easier product finding and shorter travelled distance	Focuses on contextual trolley assistance rather than changing product/list routes
Coles Smart Cart [9]	AI, sensors, scales and cart display	Store-map/navigation functions	Yes	Integrates tracking, map and checkout into the cart	Cited source does not establish adaptive shopping-list rerouting
Intermarché Smart Cart [10]	Clip-on computer-vision device	Smart-cart interface	Yes	Adds smart functions to existing carts	Adaptive route recalculation is not established in cited source
Woolworths Scan&Go [11]	Cart-mounted scanner and display	Search/list features developing	Yes	Scan and pack while shopping	Adaptive route guidance is not established in cited material
Techno M-Cart [12]	Robotic cart with sensors, gesture control, and mobile app	Autonomous movement and obstacle avoidance	No	Hands-free cart movement and obstacle detection	Does not focus on product-location guidance or adaptive shopping-list routing
Clevart	Clip-on cart interface + indoor positioning + product/location database + adaptive routing	Accurate product finding with dynamic route recalculation	Yes	Intended to reduce search time, walking distance and backtracking	Must validate positioning accuracy, product-data maintenance and route performance

Identified Gap and Clevart Contribution

Existing research shows that list-aware, map-based guidance and cart-integrated assistance can make product finding easier and improve shopping efficiency [4], [5]. Other studies have demonstrated the feasibility of indoor positioning and product localization in supermarkets [6], [7], and a more infrastructure-heavy autonomous cart has reported shorter shopping and checkout times [8]. However, reliable supermarket guidance depends on several elements working together: the shopper must be located accurately, product-location records must remain current, and the route must respond when the shopping situation changes.

Clevart will therefore focus its main engineering contribution on **adaptive in-store product guidance**. Rather than treating product-location maintenance itself as the novelty, Clevart will use reliable product-location information together with indoor shopper localization and shopping-list-based route planning. When an item is added, removed, missed, found, or when the shopper deviates from the planned route, the remaining route can be recalculated dynamically.

Clevart will later be evaluated against a simpler shopping method, such as a conventional shopping list or static store map, using measurable outcomes such as **search time, walking distance, backtracking frequency, and successfully located items**. This comparison will determine whether adaptive guidance provides a measurable improvement rather than simply adding additional digital features.

Strengths and credentials

Members	Strengths and Credentials
Ayed Sultan Al-Mutairi	Java programming, database systems/SQL, computer networks, embedded systems, and electronic circuits.
Khalid Fahad Abu-Shaibah	Java programming, computer networks, embedded systems, and electronic circuits.
Yousof Osama Dardir	Database design, software requirements, Programming/Coding, system architecture, API integration, algorithm design, software testing.
Yousef Naif Al-Otaibi	Java programming, computer networks, embedded systems, electronic circuits, and software testing.
Ali Mishal Al-Laho	machine learning, Programming, electronics, information retrieval, circuit design.

Learning Plan

The team will address its current skill gaps by assigning lead members to the main technical areas. Ayed Sultan Al-Mutairi will lead indoor positioning and the evaluation of BLE, UWB, and visual-marker techniques. Khalid Fahad Abu-Shaibah will lead the cart hardware, ESP32 integration, power system, and supporting electronics. Yousof Osama Dardir will lead the application, database/API integration, and adaptive routing algorithms. Ali Mishal Al-Laho will lead the evaluation of optional computer-vision techniques, while Yousef Naif Al-Otaibi will support hardware integration and system testing. Each lead member will study and prototype their assigned area before the final system architecture is selected.

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Appendix A: Project Selection Guideline Checklist

The following table shows how the project satisfies the project selection guidelines provided by the Computer Engineering Department at Kuwait University.

Guideline	**Satisfaction Evidence**
A tangible product should be built	**Yes.** Clevart will include a **clip-on cart-mounted unit** with a display and supporting electronics, integrated with the software system and supermarket product/location database. The prototype will demonstrate product search, item tracking, indoor positioning and adaptive route guidance.
The targeted problem is clearly stated.	**Yes.** Clevart targets the time and effort shoppers lose while searching for products, backtracking between aisles, and adjusting their route when products are missed, added, or relocated. The main engineering focus is **adaptive in-store product guidance**.
The problem is realistic and is based on users' needs.	**Yes.** Field research identified difficulties locating exact products, unnecessary aisle searching, backtracking, and dependence on employees. These findings are also consistent with related supermarket-navigation research discussed in the Background section.
There is no ambiguity in measuring the success of the final system.	**Yes.** Clevart will be compared against a baseline such as shopping using a paper shopping list or static store map. The primary metrics will be **product-search time, walking distance, number of backtracking events, and number of**

	requested items successfully found. Numerical improvement targets will be established after a small pilot test rather than being assumed in advance.
At least *4* constraints are identified and have impact on the design for the project.	**Yes. Compatibility with existing carts** leads to a removable clip-on unit rather than building a new trolley. **Cost** favors lower-cost technologies such as BLE, barcode scanning and visual markers before more expensive UWB or advanced computer vision. **Battery life** requires low-power communication and efficient hardware. **Safety/usability** favors a stationary cart-mounted guidance device with minimal user interaction rather than autonomous cart movement. For the prototype, the team will target a cart-unit hardware cost of approximately 50–70 KWD, with an additional small budget for BLE beacons or positioning equipment in the test area. Final costs will depend on the positioning technology selected during prototyping.
At least two or more APIs would be integrated with each other.	**Yes.** Firebase Authentication and Cloud Firestore can be used for user accounts, shopping lists, and product data, while Google ML Kit Barcode Scanning can be used to identify purchased items. The indoor-positioning data will also be integrated with the digital store-map and routing module, so the shopper's estimated position can be used to

	calculate and continuously update the route.
The project allows for more than one alternative design	**Yes.** Three realistic designs are available: **(1)** a cart-mounted clip-on display and sensors, **(2)** a phone-only application using the customer's smartphone, and **(3)** a hybrid system combining a cart display with a mobile application. These alternatives can be compared based on cost, positioning accuracy, usability, battery requirements, and implementation complexity.
The project can be completed as specified by the process using available resources.	**Yes.** The prototype can be built using commercially available microcontrollers, displays, barcode scanners, BLE devices/beacons, mobile/software development tools and database services. The project can begin with a small simulated supermarket layout and later be scaled to a larger environment.
The project does not significantly resemble a project that has been completed locally.	**Yes, although related KU projects exist. Groce-Tech** [13] is a grocery-shopping application using AR, self-checkout, authentication and store navigation, but it does not focus on cart-mounted adaptive multi-item route recalculation. **Kaf Kafif** [14] assists visually impaired shoppers using a smart glove, sensors, camera and navigation, while Clevart targets general supermarket shoppers and adaptive

	product-location routing. **Techno M-Cart** is a robotic cart centered on obstacle detection, gesture control and autonomous cart movement, whereas Clevart uses a standard non-robotic cart and focuses on locating products and dynamically recalculating the shopping route.
The project is different from existing systems in terms of speed, functionality, cost or effect on environment.	**Yes.** Clevart's main contribution is combining reliable product-location information, indoor shopper localization and **adaptive shopping-list route recalculation** in a lightweight attachment for existing carts. When an item is added, removed, missed or found, or when the shopper deviates from the planned path, the remaining route can be recalculated rather than requiring the shopper to manually plan a new route.

Appendix B: Meeting Minutes

Kuwait University

College of Engineering and Petroleum
Computer Engineering Department

Team Meeting Minutes #1

Team Name:	**ArchWorks**	**Date:**	**18/9/2026**
Start Time:	**19:45**	**Finish Time:**	**20:30**

Members Present:	Ayed Sultan Al-Mutairi**,** Ali Mishal Al-Laho, Yousef Naif Al-Otaibi **,** Khalid Fahad Abu-Shaibah, Yousof Osama Dardir
Members Excused:	**0**
Members Tardy:	**0**
Members Absent:	**0**

Summary of meeting

1. Decided on Team Leader
2. Discussed potential project ideas

Task list:

Team Member	Assigned Tasks	Start Date	Due Date
Ayed Sultan Al-Mutairi	Prepare new project ideas	**18/9/2026**	**24/9/2026**
Ali Mishal Al-Laho	Prepare new project ideas	**18/9/2026**	**24/9/2026**
Yousef Naif Al-Otaibi	Prepare new project ideas	**18/9/2026**	**24/9/2026**
Khalid Fahad Abu-Shaibah	Prepare new project ideas	**18/9/2026**	**24/9/2026**
Yousof Osama Dardir	Prepare new project ideas	**18/9/2026**	**24/9/2026**

Next meeting will be held on 22/9/2026

Kuwait University

College of Engineering and Petroleum
Computer Engineering Department

Team Meeting Minutes #2

Team Name:	**ArchWorks**	**Date:**	**22/9/2026**
Start Time:	**20:00**	**Finish Time:**	**21:50**

Members Present:	Ayed Sultan Al-Mutairi**,** Ali Mishal Al-Laho, Yousef Naif Al-Otaibi **,** Khalid Fahad Abu-Shaibah, Yousof Osama Dardir
Members Excused:	**0**
Members Tardy:	**0**
Members Absent:	**0**

Summary of meeting

1. Discussed potential project ideas

Task list:

Team Member	Assigned Tasks	Start Date	Due Date
Ayed Sultan Al-Mutairi	Research the proposed project idea and try to find areas of improvement	**22/9/2026**	**23/9/2026**
Ali Mishal Al-Laho	Come up with additional project ideas	**22/9/2026**	**23/9/2026**
Yousef Naif Al-Otaibi	Research the project ideas and try to measure feasibility	**22/9/2026**	**23/9/2026**
Khalid Fahad Abu-Shaibah	Improve on his project idea based on Dr. Maha's feedback	**22/9/2026**	**23/9/2026**
Yousof Osama Dardir	Format the presentation for the ideas further	**22/9/2026**	**23/9/2026**

Next meeting will be held on 26/9/2026

Kuwait University

College of Engineering and Petroleum
Computer Engineering Department

Team Meeting Minutes #3

Team Name:	**ArchWorks**	**Date:**	**26/9/2026**
Start Time:	**20:00**	**Finish Time:**	**20:40**

Members Present:	Ayed Sultan Al-Mutairi**,** Ali Mishal Al-Laho, Yousef Naif Al-Otaibi **,** Khalid Fahad Abu-Shaibah, Yousof Osama Dardir
Members Excused:	**0**
Members Tardy:	**0**
Members Absent:	**0**

Summary of meeting

1. Finished Handout 1
2. Finished Choosing your project idea – Team Worksheet
3. Presented additional project ideas

Task list:

Team Member	Assigned Tasks	Start Date	Due Date
Ayed Sultan Al-Mutairi	Look into handout 2 and Research the project ideas	**26/9/2026**	**28/9/2026**
Ali Mishal Al-Laho	Check the feasibility of one of the newly presented ideas	**26/9/2026**	**28/9/2026**
Yousef Naif Al-Otaibi	Research the project ideas	**26/9/2026**	**28/9/2026**
Khalid Fahad Abu-Shaibah	Research the project ideas	**26/9/2026**	**28/9/2026**
Yousof Osama Dardir	Research the project ideas	**26/9/2026**	**28/9/2026**

Next meeting will be held on 28/9/2026

Kuwait University

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Team Meeting Minutes #4

Team Name:	**ArchWorks**	**Date:**	**28/9/2026**
Start Time:	**12:15**	**Finish Time:**	**12:30**

Members Present:	Ayed Sultan Al-Mutairi**,** Ali Mishal Al-Laho, Yousef Naif Al-Otaibi **,** Khalid Fahad Abu-Shaibah, Yousof Osama Dardir
Members Excused:	**0**
Members Tardy:	**0**
Members Absent:	**0**

Summary of meeting

1. Abstract sections were distributed between members

Task list:

Team Member	Assigned Tasks	Start Date	Due Date
Ayed Sultan Al-Mutairi	-Prepare the "Needs" section and the Meeting Minutes -Prepare the "Background and Technology Survey" section -Add his Strengths and Credentials	**28/9/2026**	**30/9/2026**
Ali Mishal Al-Laho	-Add his Strengths and Credentials -Prepare the overall document	**28/9/2026**	**30/9/2026**
Yousef Naif Al-Otaibi	-Add his Strengths and Credentials -Prepare the Poster	**28/9/2026**	**30/9/2026**
Khalid Fahad Abu-Shaibah	-Add his Strengths and Credentials -Prepare the Presentation	**28/9/2026**	**30/9/2026**
Yousof Osama Dardir	-Finish Handouts 4 and 5 -Add his Strengths and Credentials -Prepare "Objectives" Section	**28/9/2026**	**30/9/2026**

Next meeting will be held on 3/10/2026

Kuwait University

College of Engineering and Petroleum
Computer Engineering Department

Team Meeting Minutes #5

Team Name:	**ArchWorks**	**Date:**	**3/10/2026**
Start Time:	**16:00**	**Finish Time:**	**17:00**

Members Present:	Ayed Sultan Al-Mutairi, Ali Mishal Al-Laho, Yousef Naif Al-Otaibi
Members Excused:	Khalid Fahad Abu-Shaibah, Yousof Osama Dardir
Members Tardy:	**0**
Members Absent:	**0**

Summary of meeting

1. Looked into Dr. Maha's feedback for the first abstract draft

Task list:

Team Member	Assigned Tasks	Start Date	Due Date
Ayed Sultan Al-Mutairi	Rebuild the Backgrounds section based on Dr. Maha's feedback	**3/10/2026**	**5/10/2026**
Ali Mishal Al-Laho	Rebuild the Needs section based on Dr. Maha's feedback	**3/10/2026**	**5/10/2026**
Yousef Naif Al-Otaibi	Rebuild the Objectives section based on Dr. Maha's feedback	**3/10/2026**	**5/10/2026**
Khalid Fahad Abu-Shaibah	Prepare the slides for the presentation	**3/10/2026**	**5/10/2026**
Yousof Osama Dardir	Rebuild the Appendix A section based on Dr. Maha's feedback	**3/10/2026**	**5/10/2026**

Next meeting will be held on 6/10/2026

Kuwait University

College of Engineering and Petroleum
Computer Engineering Department

Team Meeting Minutes #6

Team Name:	**ArchWorks**	**Date:**	**6/10/2026**
Start Time:	**17:00**	**Finish Time:**	**17:45**

Members Present:	Ayed Sultan Al-Mutairi, Ali Mishal Al-Laho, Khalid Fahad Abu-Shaibah, Yousof Osama Dardir
Members Excused:	Yousef Naif Al-Otaibi
Members Tardy:	**0**
Members Absent:	**0**

Summary of meeting

1. Looked into Dr. Maha's feedback for the second abstract draft
2. Distributed the slides to each member

Task list:

Team Member	Assigned Tasks	Start Date	Due Date
Ayed Sultan Al-Mutairi	Rebuild the Backgrounds section based on Dr. Maha's feedback	**6/10/2026**	**7/10/2026**
Ali Mishal Al-Laho	Rebuild the Needs section based on Dr. Maha's feedback	**6/10/2026**	**7/10/2026**
Yousef Naif Al-Otaibi	Rebuild the Objectives section based on Dr. Maha's feedback	**6/10/2026**	**7/10/2026**
Khalid Fahad Abu-Shaibah	Prepare the slides for the presentation	**6/10/2026**	**7/10/2026**
Yousof Osama Dardir	Rebuild the Appendix A section and Learning plan based on Dr. Maha's feedback	**6/10/2026**	**7/10/2026**

Next meeting will be held on 9/10/2026